SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	Hands Off Our Rights
Project / Product Name	Future Skills
Final Live Deployment URL	https://future-skills.replit.app
Codebase Repository URL	https://github.com/pekoremon/future-skills
Name (Leader)	Hong Zhi Yang
Email & phone (Leader)	zhiyang19015@gmail.com | +60 11-4035 9109

2. Project Description

[Maximum 250 words]

Future Skills is a free web app that helps Malaysian IT students choose which part of IT to aim for while AI changes which jobs are needed. 35.2% of tertiary-educated workers in Malaysia hold semi- or low-skilled jobs (DOSM, Q2 2026), and students mostly pick a specialisation by hearsay.

A student uploads a CV (optional), answers 36 interest questions, then takes a short skills test on only the 3–5 IT areas these point to. The app then:

1. ranks 25 IT roles in 5 areas with a score split into area fit, interest, skills and trend, and lists the skills still missing;

2. shows a Future Skills Radar from TalentCorp MyCOL, DOSM/OpenDOSM and arXiv AI-paper counts, each with source and date, refreshable live with “Update data”;

3. builds an ordered path of checked free courses and videos, with hours and weeks at the student's own study time, plus HRD Corp-claimable options;

4. tags every role and skill to UN SDG targets with a reason; picked goals reorder the student's jobs;

5. compares three future scenarios (current trends, fast AI adoption, Malaysian demand leads), labelled as estimates;

6. lets anyone change the trend data (“What if”) and watch each step of the result change.

An LLM (GPT-5.4 mini via KrackedDevs) only reads CVs and words explanations; our own sourced tables decide every number, and a reply with an invented number is rejected. The app works in English, Bahasa Melayu and Chinese, on phones, with no account.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

One Python program, one browser app, sourced tables, one guarded LLM.

User → front end → JSON API → scoring engines → data tables ← trend pipeline; LLM only for wording; hosted on Replit.

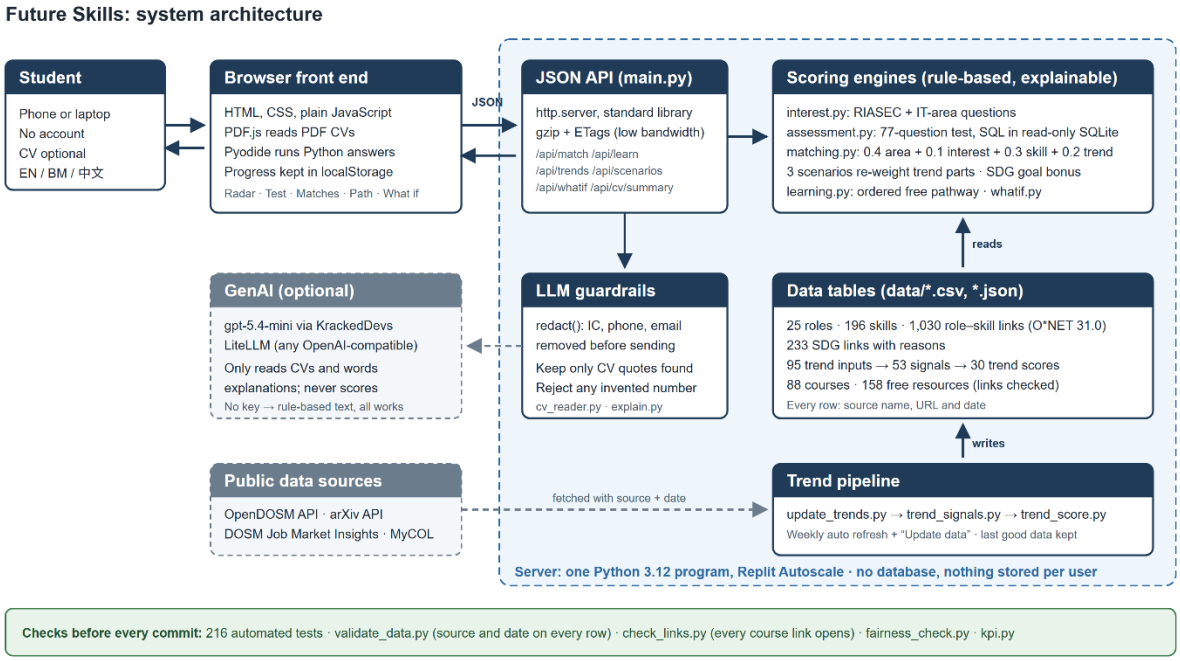


Figure 1. System architecture (also docs/architecture.png in the repository).

1. Front end (src/static/): HTML, CSS and plain JavaScript, no framework or build step. English / Bahasa Melayu / Chinese menu, light/dark mode, 40 px touch targets for phones. PDF CVs are read in the browser with PDF.js; Python test answers run in the browser with Pyodide, so no student code runs on our server. Answers and learning progress stay in the browser (localStorage); there are no accounts.

2. Back end (src/main.py): a JSON API on the Python 3.12 standard library (http.server), no packages to install. Answers are gzip-compressed with ETags for low-bandwidth use. Endpoints: /api/questionnaire, /api/assessment/*, /api/cv/summary, /api/match, /api/learn, /api/explain, /api/trends, /api/trends/update, /api/scenarios, /api/whatif.

3. Scoring engines (scripts/): interest.py (RIASEC + IT-area questions), assessment.py (77-question IT skills test; SQL answers run in a read-only in-memory SQLite database), matching.py (area fit, then role score = 0.4 area fit + 0.1 interest + 0.3 skill + 0.2 trend, with each part and the skill gap returned), learning.py (ordered pathway), whatif.py (re-runs the whole pipeline on a private copy of changed trend data).

4. Data layer (data/), our database: CSV/JSON tables where every row carries source_name, source_url and source_date, checked by validate_data.py: 25 roles and 196 skills with 1,030 role–skill links (O*NET 31.0), 233 SDG

links each with a one-line reason, 95 dated trend inputs → 53 signals → 30 trend scores, 88 courses and 158 free resources with the date each link was checked (check_links.py).

5. Trend pipeline: update_trends.py fetches the OpenDOSM API (employment, wages) and the arXiv API (AI papers per IT area), reads DOSM Job Market Insights and MyCOL 2024/2025; trend_signals.py gives each signal a direction, strength and confidence; trend_score.py turns them into 0–100 scores. If a source fails, its previous rows are kept (last good snapshot).

6. GenAI: any OpenAI-compatible chat API; live: gpt-5.4-mini through the SDC sponsor KrackedDevs LiteLLM gateway (gpt-oss-120b on Fireworks AI until 31 Oct). cv_reader.py asks for quoted CV evidence per IT area and keeps only quotes really in the CV; explain.py words one match's computed facts and rejects any reply containing a number not in them. Without a key, rule-based text is shown and every feature still works.

7. Hosting and quality: a Replit Autoscale Deployment with the key in Replit Secrets; GitHub for code. 216 automated tests, the data validator and the link checker run before each commit.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

Phase 2 upgrading (28–30 Oct): Bahasa Melayu and Chinese interface; a "What if the trend data changed?" page that re-runs signals → scores → ranking on a private copy for the live change test.

48-hour physical sprint (31 Oct – 1 Nov, over 100 commits: code commits and Replit publishes, all deployed):

• **Trend radar as the landing page**, with an "Update data" button that fetches OpenDOSM and arXiv live, and links to every input with its source and date.

• **New trend score (v3):** Malaysian demand uses the share of an area's roles on MyCOL instead of one thin DOSM series; AI momentum is a continuous arXiv growth scale. We dropped IT Management (no Malaysian signal) and added Statisticians and Special Effects Artists: 5 areas, 25 roles.

• **Pathways:** 175 more checked free courses and videos; pathway coverage rose to 86.9% of needed skills (target 40%); a link checker; progress tracking without accounts.

• **Matches page:** scenario switch, score bars, a short AI "Why this job?" with guardrails; students pick the UN goals they care about, and jobs supporting one get 5 extra points and move up.

• **Positioning:** the Trends page compares us with JobStreet, Hiredly, LinkedIn, MyFutureJobs, MyMahir, MyCOL, Coursera, Udemy and edX.

• **Trust and UX:** skills-test score ranges, a CV privacy box (our code removes IC numbers, phones and emails before the LLM), light/dark mode, gzip for slow connections; a fairness check (one CV sent as 8 different students gets identical scores).

• **Live LLM moved to the sponsor KrackedDevs API** (gpt-5.4-mini; any OpenAI-compatible API works) and kpi.py to measure our KPIs from the data.

Problem-solving: testers abandoned the full 24-question test, so the flow became CV → interests → a focused test. DOSM publishes job ads for only one IT occupation, so the old trend score barely separated areas; MyCOL coverage made it Malaysian and explainable.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	Your Response
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	**KPI 2 — Pathway coverage (share of needed skills with at least one free, manually checked course or video):** Measured the same way at both points by scripts/kpi.py (a free course or free video whose link we checked): Phase 1 baseline 29.0% (54 of 186 skills, commit 6802b63) → Phase 2 prototype 86.9% (166 of 191 skills), above the 40% target. By the proposal's narrower count (courses.csv only): 14.0% (26 of 186) → 43.5% (83 of 191), also above target. **KPI 1 — Career clarity (testers who can name a target role, its top missing skills and a next course after one session):** target ≥70% of ≥20 IT students. Result: **20 testers: 35% before → 75% after (7 → 15 of 20); confidence 3.3 → 3.7 of 5** (pilot at the physical hackathon, 31 Oct – 1 Nov 2026). Target met: at least 70% of at least 20 testers. After the session, 17 of 20 (85%) said the results changed their plan (11 "a lot"), 12 of 20 (60%) said they will start the course they named (8 "maybe"), and testers rating their confidence 4 or 5 rose from 10 to 13. **Domain must-haves now measured:** 25 of 25 roles and 196 of 196 skills tagged to an SDG target with a reason; 95 of 95 trend inputs carry a source and fetch date; 3 scenarios change rankings (e.g. demo profile Aiman: Blockchain Engineers rises from #3 to #2 under fast AI adoption). Fairness: the same CV sent as 8 students who differ only in name, gender, age, ethnicity, religion and home state gets identical area and role scores. Picking a UN goal changes the ranking: for Aiman, choosing SDG 17 moves Data Scientists from #5 to #1.
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	• **Code counters, reproducible by anyone:** python scripts/kpi.py (now) and python scripts/kpi.py --rev 6802b63 (the Phase 1 submission commit) compute coverage from the data files themselves. • **Data validator** (validate_data.py: every row has source and date), 216 automated tests (python -m unittest discover tests) and the fairness check (python scripts/fairness_check.py), all run on the submitted commit. • **Link checker** (check_links.py) opens every course, video and certification link.

Metric Category	Your Response
	• **Live API:** POST /api/match on the deployed app returns each score's parts (Section 4.1). • KPI 1: the pilot's Google Form asks the same 3 questions before and after one session; python scripts/clarity_kpi.py responses.csv counts a tester as clear when a job, at least 2 skills and a course are named, and prints each row for a hand check. Form: https://docs.google.com/forms/d/e/1FAIpQLSemWTTx0Gzn7CJ4igIBwnOeNsCOR4qJPu7kVO80mszVF2fy4A/viewform Responses: https://docs.google.com/spreadsheets/d/1ZorKD7hgimQIG9NURNKgt8dAjFOnagovdUyOE2hS7q4/edit
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	Malaysia's 20 public universities, polytechnics and TVET colleges run career centres that advise thousands of IT students a year with static guides. Future Skills gives them a free, explainable tool they can hand out by link: no install, no student accounts, low bandwidth, three languages. Each recommendation cites MyCOL, DOSM or O*NET, so a counsellor can check it. HRD Corp-claimable programmes are listed separately for working adults reskilling into IT, supporting SDG 4.4 and 8.5. Because the app collects no age, gender or background, recommendations depend only on interests, skills and the CV.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	• Hosting: one small Python server, no database, on a Replit Autoscale Deployment (US$1 a month plus US$0.40 per million requests, Oct 2026): about RM 5–25 a month for 5,000 students. • Trend freshness costs RM 0: OpenDOSM and arXiv are free APIs, and the server re-fetches them by itself once the data is 7 days old (AUTO_UPDATE_DAYS); “Update data” refreshes on demand. DOSM Job Market Insights (quarterly) and MyCOL (yearly) take about 1 hour to add per release. • The only paid part is the optional LLM (gpt-5.4-mini, US$0.75 / 4.50 per million tokens in / out): 2 calls, about 2,400 tokens in and 1,200 out per student ≈ RM 0.03 (at most RM 0.12); 5,000 students ≈ RM 150 a year. The app works fully without it. • Packaged as a JSON API (/api/match, /api/learn, /api/trends), so a university portal or HR system can embed it. • Model: free for students; paid white-label or API licence for career centres, training providers and HR teams.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

1. Match score (scripts/matching.py) — decomposable, every part shown on screen

total = 0.4 × area_fit + 0.1 × interest + 0.3 × skill + 0.2 × trend (missing parts are left out and the others rescaled, never filled)

2. Trend score (scripts/trend_score.py)

part = 50 + 50 × Σ(confidence × strength) / Σ confidence confidence: high 1, medium 2/3, low 1/3; strength −1…+1

trend = w_demand × Malaysian demand + w_ai × AI momentum current trends 0.5/0.5 · fast AI 0.2/0.8 · Malaysian demand leads 0.8/0.2

3. Live log — POST https://future-skills.replit.app/api/match {"profile_id":"P01"} (fictional student Aiman, data computed 2026-10-31)

IT02 Computer Programmers area 77.4→31.0 interest 98.1→9.8 skill 63.8→19.1 trend 85.3→17.1 = 77.0

IT01 Software Developers area 77.4→31.0 interest 99.4→9.9 skill 62.3→18.7 trend 85.3→17.1 = 76.7

IT04 Blockchain Engineers area 77.4→31.0 interest 95.1→9.5 skill 66.4→19.9 trend 77.0→15.4 = 75.8

Check: IT02 trend = 0.5 × 83.3 (demand) + 0.5 × 87.3 (AI) = 85.3. Under fast AI, IT04 trend = 0.2 × 66.7 + 0.8 × 87.3 = 83.2, so Blockchain Engineers moves from rank 3 to 2 and Software Developers drops to 3: the scenario changes parameters, not hand-written outputs.

4. KPI counter — python scripts/kpi.py

Data: commit 6802b63 (Phase 1) → Pathway coverage: 54 of 186 needed skills have a free, checked course = 29.0%

Data: data/ (working tree) → Pathway coverage: 166 of 191 needed skills have a free, checked course = 86.9% (Phase 2 target 40.0%)

SDG tags: 25 of 25 roles, 196 of 196 skills

5. Guardrail (scripts/explain.py): every number in the LLM reply must appear in the computed facts sent to it; otherwise the reply is discarded and a rule-based summary is shown (source: "fallback").

6. KPI 1 counter — python scripts/clarity_kpi.py responses.csv (the pilot's Google Form answers, before and after one session)

Career clarity: 20 testers; 35% before → 75% after (7 → 15 of 20)

Confidence (1-5): 3.3 before → 3.7 after

Target: ≥ 70% of ≥ 20 testers → met

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Who first: IT students at UTAR and partner universities through their career centres, then polytechnics and TVET colleges; a realistic first year is 3–5 institutions and about 5,000 students.

With funding we would:

1. Automate freshness: the weekly OpenDOSM and arXiv refresh already runs; we would keep each snapshot (not only the latest) in a managed database so changes over time become signals, and add each new DOSM Job Market Insights quarter and MyCOL edition on release.
2. **Add volume data:** job-ad counts per skill from Malaysian job boards or MyMahir, under data-sharing agreements, to turn area-level signals into rising and declining skills.
3. **Scale the platform:** move from one Replit instance to a containerised API with caching; scoring is pure arithmetic over small tables, so one server can serve thousands of students per hour, and LLM calls stay optional and capped per user.
4. **Partner channels:** Rotary and XMUM networks for pilots; HRD Corp-registered training providers to verify claimable courses; employers for an HR view that shows a team's skills-gap heatmap.
5. **Validate and stay fair:** run the career-clarity study at each partner, calibrate the weights with counsellors, repeat the fairness check on the LLM CV reading with real CVs, and comply with Malaysia's PDPA (IC, phone and email removed before the LLM, no CV storage, consent screens).

Sustainability: free for students; a yearly licence for institutions and HR teams covers hosting and data upkeep.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- [x] The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- [x] The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- [x] The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- [x] The final pitch deck has been exported and saved as a flattened PDF file.
- [x] All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET 31.0 Database (US DOL/ETA)	Dataset	CC BY 4.0
DOSM Labour Market Review, Job Market Insights, Graduates Statistics 2024	Dataset	DOSM release; cited “Source: DOSM”
OpenDOSM API (productivity_qtr, wages_occupation)	Dataset (API)	CC BY 4.0
TalentCorp MyCOL 2024/2025	Dataset (publication)	Public publication, cited
arXiv API (paper counts only)	Dataset (API)	Metadata CC0; arXiv API terms
UN SDG Global Indicator Framework; UN SDG icons	Dataset; images	UN public document; UN SDG icon guidelines
Free course/video links (freeCodeCamp, CS50, Microsoft Learn, Cisco NetAcad, etc.)	Linked content	© each provider; linked, not copied

gpt-5.4-mini (OpenAI) via KrackedDevs LiteLLM gateway (live)	Pre-trained model (API)	OpenAI usage policies; KrackedDevs sponsor terms
gpt-oss-120b (OpenAI) via Fireworks AI (until 31 Oct)	Pre-trained model	Apache 2.0; Fireworks AI terms
Pyodide 0.26.4	Code (browser library)	MPL 2.0
PDF.js 3.11.174 (Mozilla)	Code (browser library)	Apache 2.0
Python 3.12 standard library, SQLite	Code	PSF License; public domain
Unsplash photos (images.unsplash.com)	Images	Unsplash License
Google Fonts: Press Start 2P, Orbitron, Pirata One	Code (web fonts)	SIL Open Font License 1.1
Google Forms and Sheets (pilot survey for KPI 1)	Tool	Google Terms of Service

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

Team leader: Hong Zhi Yang Signature:

Date: 1 November 2026